

BEATRIZ SOUZA

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EDUCATION

APR 2023-now	University of Stuttgart , Germany PhD in Computer Science Advisor: Michael Pradel Research Area: AI4Code, Execution, Program Analysis
OCT 2021-FEB 2023	Federal University of Pernambuco , Brazil Master's in Computer Science Advisor: Marcelo d'Amorim Thesis: Learning to Detect Text-Code Inconsistencies with Weak and Manual Supervision
MAY 2017-JUNE 2021	Federal University of Campina Grande , Brazil Bachelor's in Computer Science Advisor: Rohit Gheyi Thesis: Most Higher Order Mutants are Useless for Method Level Operators
MAY 2013-SEPT 2016	Federal Institute of Education, Science and Technology of Paraíba , Brazil High School with a Technical Degree in Computer Science Advisor: Gustavo Vieira

WORK EXPERIENCE

2026 (JUNE-AUG)	Microsoft , Redmond, WA, USA - APPLIED SCIENCES INTERN Mentor: Ashish Tiwari - PROSE Project: Learning-guided early failure prediction for AI model escalation.
2024 (APR-JULY)	Microsoft Research , Redmond, WA, USA - RESEARCH INTERN Mentor: Suman Nath - Cloud Systems Reliability Project: <i>runtime monitoring</i> of distributed systems.
2022 (JULY-DEC)	Software Lab , Stuttgart, Germany - RESEARCH INTERN Advisor: Michael Pradel Project: <i>LExecutor</i> , a learning-guided approach for executing arbitrary code.
MAY 2020-FEB 2021	Analytics Lab , Campina Grande, Brazil - SOFTWARE DEVELOPER Mentors: João Brunet and Nazareno Andrade Projects: <i>Tá de Pé?</i> and <i>Monitor Cidadão</i> , web applications to identify risks in contracts.
NOV 2019-OCT 2020	UFCG , Campina Grande, Brazil - RESEARCH ASSISTANT Advisor: Rohit Gheyi Project: Identifying useless mutants using Z3.
AUG 2018-JULY 2019	SPLAB , Campina Grande, Brazil - RESEARCH ASSISTANT Advisor: Patrícia Machado Project: Investigating the effectiveness of automated test generators.
JULY 2016-SEPT 2016	Papagaio , João Pessoa, Brazil - WEB DEVELOPER INTERN Mentor: Gustavo Vieira Project: Development of an e-commerce platform.
AUG 2015-JULY 2016	IFPB , Cajazeiras, Brazil - RESEARCH ASSISTANT Advisors: Gustavo Vieira and Wilza Moreira Project: <i>Segundo Mendel</i> , a mobile application to learn genetic concepts.

SERVICE

- Student volunteer: ASE 2020 and ICSE 2026.

- PREPRINT'26 **FlyCatcher: Neural Inference of Runtime Checkers from Tests.**
Beatriz Souza, Chang Lou, Suman Nath, Michael Pradel.
- ICSE'25 **Treefix: Enabling Execution with a Tree of Prefixes.**
Beatriz Souza and Michael Pradel.
International Conference on Software Engineering
- FSE'25 **ChangeGuard: Validating Code Changes via Pairwise Learning-Guided Execution.**
Lars Gröninger, **Beatriz Souza** and Michael Pradel.
Symposium on the Foundations of Software Engineering
- FSE'23 **LExecutor: Learning-Guided Execution.**
Beatriz Souza and Michael Pradel.
Symposium on the Foundations of Software Engineering
- PREPRINT'22 **Code Generation Tools (Almost) for Free?**
A Study of Few-Shot, Pre-Trained Language Models on Code.
Patrick Bareiß, **Beatriz Souza**, Marcelo d'Amorim, Michael Pradel.
- IST'21 **Identifying Method-Level Mutation Subsumption Relations using Z3.**
Rohit Gheyi, Márcio Ribeiro, **Beatriz Souza**, Marcio Guimarães, Leo Fernandes, Marcelo d'Amorim, Vander Alves, Leopoldo Teixeira, Balduino Fonseca.
Elsevier Information and Software Technology
- SBES'20 **A Large Scale Study On the Effectiveness of Manual and Automatic Unit Test Generation.**
Beatriz Souza and Patrícia Machado.
Brazilian Symposium on Software Engineering
- SBES'20 **A Lightweight Technique to Identify Equivalent Mutants.**
Beatriz Souza and Rohit Gheyi.
Brazilian Symposium on Software Engineering
- ASE'20 **Identifying Mutation Subsumption Relations.**
Beatriz Souza.
International Conference on Automated Software Engineering
- SPLASH'19 **Is Mutation Score a Fair Metric?.**
Beatriz Souza.
International Conference on Systems, Programming, Languages, and Applications: Software for Humanity

AWARDS AND HONORS

- **Selected for the 12th Heidelberg Laureate Forum as a young researcher.**
- **ACM SIGSOFT Distinguished Paper Award at The Symposium on the Foundations of Software Engineering (FSE'23)** for the work *LExecutor: Learning-Guided Execution*.
- **Best paper at The Brazilian Symposium on Software Engineering (SBES'20)** for the work *A Large Scale Study On the Effectiveness of Manual and Automatic Unit Test Generation*.
- **First Place at The Undergraduate Research on Software Engineering Competition (SBES'20)** for the work *A Lightweight Technique to Identify Equivalent Mutants*.
- **Third Place at The ACM Student Research Competition (ASE'20)** for the work *Identifying Mutation Subsumption Relations*.
- **Third Place at The ACM Student Research Competition (SPLASH'19)** for the work *Is Mutation Score a Fair Metric?.*

Bachelor theses

- Aleksis Vezenkov. Performance-Improving Refactorings for Python. 2025

Master theses

- Lars Gröninger. Reasoning about Code Changes via Pairwise Learning-Guided Execution. 2024

TEACHING EXPERIENCE

APR 2025-SEPT 2025	Teaching assistant at SOLA, Stuttgart, Germany
OCT 2024-MAR 2025	Programming Paradigms - I created and graded exercise lists and addressed
APR 2023-SEPT 2023	questions for a class of 80+ students.
MAR 2018-AUG 2018	Teaching assistant at UFCG, Campina Grande, Brazil
	Mathematical Foundations for Computer Science - I helped professor by grad-
	ing exercise lists and addressing questions for a class of 46 students.
JULY 2015-DEC 2015	Teaching assistant at IFPB, Cajazeiras, Brazil
	Competitive Programming - I collaborated in a course to prepare high school
	and undergraduate students for programming contests using C as program-
	ming language.

TALKS

Learning-guided Early Failure Prediction for AI Model Escalation

- Microsoft Office Product Group. Aug, 2026.

Keep Running: Towards AI-Assisted Approaches for Enabling and Monitoring Program Execution

- Max Planck Institute for Software Systems. September, 2025.
- 12th Heidelberg Laureate Forum. September, 2025.

FlyCatcher: Neural Inference of Runtime Checkers from Tests

- Microsoft Research Redmond. July, 2024

Learning-Guided Execution

- ELLIS Unit Stuttgart. July, 2025
- Microsoft Research Redmond. May, 2024